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To cite this article: Clayton Wukich (2020): Government Social Media Engagement Strategies and Public Roles, Public Performance & Management Review, DOI: [10.1080/15309576.2020.1851266](https://doi.org/10.1080/15309576.2020.1851266)

To link to this article: <https://doi.org/10.1080/15309576.2020.1851266>



Published online: 17 Dec 2020.



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Government Social Media Engagement Strategies and Public Roles

Clayton Wukich 

Cleveland State University

ABSTRACT

Through social media, public officials share information with the people they serve. Related scholarship has centered on open government goals such as transparency, participation, and collaboration to characterize such efforts, relying on the directional flow of information between government and community members, as opposed to the underlying message content as illustrated by public roles (i.e. public managers' view of residents as customers, partners, or citizens). Hurricane Florence provides a critical case to examine how cities in three U.S. states leveraged public roles for engagement purposes. Content analysis of Facebook data from 62 cities reveals their customer and partner-oriented focus as evidenced by frequent customer service updates and coproduction prompts. Citizen involvement efforts were far less common; few cities convened conversations about their exposure to risk or discussed how to build stronger communities. In all, this article clarifies engagement opportunities based on public roles and more explicitly tethers social media content to foundational concepts in public administration.

KEYWORDS

Social media; public roles; customers; coproduction; citizen involvement efforts

Social media applications provide government officials with channels to communicate with the public. Scholars rely on open government goals (e.g., transparency about agency activities, public participation in policy-making, and collaboration with other organizations) to categorize subsequent content (Criado et al., 2013; Ganapati & Reddick, 2014; Meijer & Thaens, 2013; Mergel, 2012). While agencies often disseminate information about their activities (i.e. transparency), (Bonsón et al., 2019; Kavanaugh et al., 2012), most fail to achieve substantive participation and collaboration (Mergel, 2016; Oliveira & Welch, 2013; Zavattaro & Brainard, 2019).

When assessing whether participation takes place, the flow of information and whether officials directly interact with community members tend to be a focal point for many scholars as opposed to the underlying message content (Brainard & Edlins, 2015; Meijer & Thaens, 2013; Mergel, 2012, 2017;

Reddick & Norris, 2013). This structural approach overlooks some fundamental considerations about the nature of government-to-citizen content, for public managers may view the people they serve through different lenses—(a) as customers who consume public goods and services, (b) as partners who coproduce those goods and services, as well as (c) citizens they deliberate with on important questions about the overall direction of government (Thomas, 2012, 2013). Such assumptions about public roles influence the social media content officials create and the engagement strategies they choose. While scholars and practitioners have explored different aspects of these roles in relation to social media, specifically participatory approaches (Bryer, 2013a; Ganapati & Reddick, 2014; Kagarise & Zavattaro, 2017; Mergel, 2012, 2017), the extent to which public managers enact, balance, and blend them remains unclear.

This article applies Thomas's (2012, 2013) public roles framework to obtain a more complete understanding of city social media engagement strategies and to contribute to scholarship focused on the structure of information exchange rather than underlying content. It demonstrates how cities enact tactics during urgent, changing conditions while providing a steady voice that sustains strategic messaging, particularly regarding service delivery. Two research questions guide this effort:

- How do city officials on social media engage the people they serve during a disaster with specific content—as customers, partners, or citizens?
- To what extent, if any, do they balance or blend these public role categories as they adjust to the changing demands of their external environments?

Content analysis of Facebook posts from 62 cities in North Carolina, South Carolina, and Virginia during Hurricane Florence in 2018 exemplifies message types that align with Thomas's (2012, 2013) public role categories. A disaster provides an appropriate case study because the urgency of the situation compels authorities to provide different types of information (Tierney et al., 2001) and in the aftermath potentially redirects public attention to policy matters such as disaster mitigation, preparation, or recovery (Birkland, 1997). Despite the opportunity for citizen participation (i.e. public involvement in policymaking), findings demonstrate the prevalence of customer and partner-orientations and how authorities balanced and at times blended them. While some city authorities occasionally engaged residents more directly as partners by soliciting feedback related to ongoing operations, they rarely facilitated participation in their policymaking processes. Such findings reaffirm calls for practitioners to more substantively engage residents whether during times of crisis (Stern, 2017) or during

more normal operating conditions (Bryer, 2013a; Mergel, 2012; Zavattaro & Brainard, 2019). The discussion concludes with practitioner recommendations and suggestions for future research.

Open government goals and social media enactment

In 2009, the Obama administration laid out the goals of their Open Government Initiative—transparency, participation, and collaboration. Agencies throughout the federal government were instructed to (a) inform the public about what they were doing (i.e. transparency), (b) encourage the public to “participate in policymaking and to provide ... the benefits of their collective expertise and information,” and (c) collaborate with other organizations (Obama, 2009, p. 1). Social media applications, which Bryer and Zavattaro (2011) define as “technologies that facilitate social interaction, make possible collaboration, and enable deliberation across stakeholders” (p. 327), provided channels to facilitate those aims. Scholars label such engagement efforts as *Government 2.0* (Mergel, 2012).

While open government goals are promoted throughout the United States and beyond, authorities focus more on transparency (or information delivery) rather than substantive, dialogic interactions related to participation or collaboration (Mergel, 2012, 2013, 2017). For example, Hand and Ching (2011) reported this to be the case with municipalities in the Phoenix metropolitan area. Because of the widespread lack of dialogue (see their survey of U.S. local government managers), Reddick and Norris (2013) went as far as to re-label such one-way practices as *Government 1.5*, and to their point, scholarship continues to demonstrate that participation is neither frequently realized in practice (Brainard & Edlins, 2015; Mossberger et al., 2013; Zavattaro & Brainard, 2019) nor strongly imbued as a value within the organizations charged with pursuing it (Piotrowski et al., 2018).

Low participation assessments are largely derived from measures such as (a) the number of reactions or replies between authorities and residents (see Bonsón et al., 2017; Brainard & Edlins, 2015; Kagarise & Zavattaro, 2017) and relatedly (b) the structure of information flow between them (see Mergel [2013, 2017] one-to-many, one-to-one, and many-to-many model). While emphasis on the structure of interactions offers useful insights, it potentially discounts the underlying content of such interactions and why organizations disseminate messages in the first place.

What types of content do organizations disseminate via social media? Not all enact the same practices (Bryer, 2013a; Meijer, 2015; Oliveira & Welch, 2013). Mergel (2016) characterizes content generation as “highly context specific” (p. 142). Organizational missions, priorities, and tasks—and the

extent to which public managers perceive social media as complementary to those things—provide elements of such context (Feeney & Welch, 2016; Mergel, 2012, 2016). Moreover, public managers' beliefs about their role in facilitating participation is germane (Bryer, 2011; Zhang & Feeney, 2018). Meijer (2015) explains that public managers justify their organization's commitment to social media with a corresponding public value (i.e. efficiency, effectiveness, or democratic norms). Therefore, understanding what public managers value and how they perceive the public they serve provides the foundation to understand how they go about developing their social media content.

How public managers approach the people they serve

The question of how public managers view and approach the people they serve is foundational to public administration (Bryson et al., 2014; Frederickson, 1991). Practitioners hold various views that include the public (a) as customers to respond to and satisfy (Osborne & Gaebler, 1992); (b) as partners in coproducing goods and services (Nabatchi et al., 2017); and (c) as citizens to empower through dialogue (Denhardt & Denhardt, 2000; Frederickson, 1982). Thomas (2012, 2013) consolidates the approaches into a three-pronged framework of public roles that consist of customer, partner, and citizen. Others offer additional categories such as interest-group advocates and voters (Frederickson, 1991; Roberts, 2004; Vigoda, 2002), but Thomas (2012, 2013) more succinctly describes the basic relationships between public managers and community members, particularly at the local level. Therefore, his framework offers a useful point of reference.

Public roles are not mutually exclusive, although normative arguments may prioritize one over another (e.g., the customer vs. citizen debate) (see Alford, 2009). Personnel may employ each simultaneously “as when people expect to be treated courteously and helpfully, like customers, as they also voice opinions, as citizens, on the nature of public programs in which they might assist, as partners” (Thomas, 2013, p. 786). Because public managers are accountable for creating and supporting such relationships (see Bryer, 2009; Feldman & Khademian, 2002; Frederickson, 1982), they ought to be able to balance and blend approaches based on the situation and the task at hand. Differentiating approaches and providing examples help to connect the framework to social media use in practice.

As customers

In the 1980s and 1990s, government reformers viewed the public as customers to satisfy; they emphasized customer-service practices from the

private sector (Barzelay, 1992; Osborne & Gaebler, 1992). From an e-government perspective, web-based efforts later emerged to make service delivery more convenient and included (a) readily available information, (b) citizen-initiated request portals, and (c) transaction-based services—things like filing taxes, renewing drivers' licenses, filing permits, or paying fines online (Moon, 2002; Reddick, 2005). Moreover, 311 call centers—some with website functions and smartphone applications—were created to provide a one-stop shop in some cities for community members to ask questions and lodge complaints (Thomas, 2012).

Through social media, content that markets services takes on a customer orientation (see DePaula et al., 2018). Letting people know where, when, and how to access services provides examples that include transportation, public safety, sanitation, and community events and activities (Bonsón et al., 2019; Criado & Villodre, 2020; Kavanaugh et al., 2012). Describing services in more detail and how they are delivered might draw additional attention. Moreover, authorities can request feedback on service quality (DePaula et al., 2018; Oliveira & Welch, 2013) or respond to customer service questions and requests (Brainard & Edlins, 2015; Criado & Villodre, 2020; Kagarise & Zavattaro, 2017; Mergel, 2013). Such efforts require considerable time and attention from the personnel who enact them.

Disasters magnify customer service challenges because they disrupt normal operations and create urgent needs within the community (see Comfort et al., 2010; Tierney et al., 2001). Conditions of uncertainty characterize such situations, therefore, authorities may not have ready answers regarding the timing or availability of services. Following their 2011 earthquakes, a New Zealand official remarked, “Especially in the beginning we had no answer to provide to people and that was of course not satisfactory” (Tagliacozzo, 2018, p. 7). Relatedly, the public expectation of omnipresence exacerbates community dissatisfaction; this is the idea that people demand instant access to officials on social media based on their past private sector interactions with customer service representatives (Zavattaro & Sementelli, 2014). However, the capacity of local government agencies may fall short of expectations because they lack personnel or are unwilling to engage directly (Hughes et al., 2014; Plotnick & Hiltz, 2016). Still, when pertinent information about service delays or event cancellations becomes available, it can be relayed to residents through social media (Panagiotopoulos et al., 2016; Sutton et al., 2014).

During nonthreat periods, some city officials take cues from corporate marketing practices to promote services for residents and development opportunities for potential investors (Kagarise & Zavattaro, 2017; Zavattaro, 2013). Strategic branding is employed (see Zavattaro, 2016), including fan-based content such as National Park photos or NASA videos

(Mergel, 2016). Some organizations employ social media to rehabilitate their public images as the Boston police department did in response to pronounced public criticism (Meijer & Thaens, 2013).

Within this context, DePaula et al. (2018) describe favorable self-presentation efforts through which authorities report “positive activity performed by the department, with positive imagery or self-referential language of gratitude (e.g., ‘we have the best,’ ‘we accomplished so much this year’)” (p. 102). The goal is to present authorities as likable, competent, or worthy of their positions. The information is not inaccurate per se; local governments have positive stories to tell. However, social media managers strategically create the content to present their organizations in a favorable light and choose to omit less favorable content. The tactic aligns with a customer orientation, especially when authorities attempt to increase public satisfaction with officials and city operations.

As Thomas (2013) suggested, the lines between categories can blur; these are not mutually exclusive viewpoints. For example, San Antonio, Texas promoted its recycling program—a public service—by instructing residents on how to effectively participate (e.g., separating certain types of recyclables) (Reddick et al., 2017). Therefore, messages both promoted the service and engaged the public as partners to improve its efficacy.

As partners

Scholars refer to coproduction as the process in which both government providers and users are involved in the delivery of a public service (Nabatchi et al., 2017). This model of interaction deviates from notions of government as producer and citizens as consumer and offers an alternative in which public involvement can determine both program outputs and outcomes (Nabatchi et al., 2017). For example—and related to Reddick et al. (2017) case—Thomas (2013) points out, “Municipal recycling programs succeed to the extent the residents voluntarily separate recyclables from other waste” (p. 85).

Scholars have yet to settle all conceptual boundaries. The degree to which participants coordinate with government represents a disputed point. On one hand, Bovaird (2007) insists that regular, long-term relationships are required. On the other, Alford (2009) allows for people to act independently as long as their behavior is “prompted by some action of the agency” (p. 23). According to Alford’s line of thinking, coproduction does not require government involvement beyond initial prompts. Therefore, public information campaigns designed to influence the public’s behavior offer a path toward coproduction without necessarily engaging the public in dialogue. This applies to emergency management in which authorities have

long sought to influence public behavior via risk reduction messages (see Sutton et al., 2014; Tierney et al., 2001).

Scholarship describes social media campaigns that inform communities on what to expect from specific hazards (e.g., hurricanes, earthquakes, and wildfires) and how to prepare households and organizations within their jurisdictions (Wukich, 2016, 2019). During disasters, specific guidance on what to do constitutes coproduction prompts as well. The goal is for residents to comply with the authorities' advice and take protective actions to reduce their exposure to risk. Sutton et al. (2014) prescribe specific message characteristics to increase audience compliance such as (a) a description of the impending hazard and likely consequences, (b) the location of those affected, (c) explicit guidance on what people should do, and (d) a timeline for those actions (i.e. when people should act) (p. 769).

Community members who comply with the messages—both during periods of threats and nonthreat periods—are more likely to remain safe during a disaster and are less likely to require assistance (Tierney et al., 2001). That assistance (i.e. personnel and resources) can then be directed to other areas, thus strengthening overall response efforts. This is important because such activities contribute to the coproduction of disaster resilience, the community's ability to proactively adapt and recover from extreme events (Comfort et al., 2010). Furthermore, information on the location, nature, and changing conditions of an incident, including its consequences (i.e. hazard impact) informs people's situational awareness and subsequent decision-making (Tierney et al., 2001). Through such information, authorities further empower residents as partners to reduce communitywide risk.

On social media, information can flow both ways. During times of threat, authorities may request information to better understand community needs and inform operational decision making (i.e. intelligence gathering) (Wukich, 2016) or actively monitor content to respond to community-initiated requests (Hughes et al., 2014). Government-citizen interactions provide responders with actionable intelligence to better help their communities. Additionally, social media enable citizen-to-citizen interactions that, in turn, can lead to coproduction (see Meijer, 2011). During disasters, the public can pass along warnings and guidance from authorities to their social networks and thus extend the reach of that important information (Robinson et al., 2019). This propagation is, in itself, coproduction because residents assist in message dissemination (Chatfield & Reddick, 2018; Criado et al., 2020).

As citizens

Citizen participation engages people in deciding the direction and decisions of government—in other words, policymaking (Nabatchi, 2012). Public

managers design, initiate, and implement such activities and citizens are then empowered as stakeholders (Bryson et al., 2014; Denhardt & Denhardt, 2000; Frederickson, 1982). Some officials engage residents more than others (Bryer, 2009, 2013a) through what Yang and Callahan (2007) label as citizen involvement efforts or “activities initiated by government to encourage citizen participation” (p. 249). Examples include public hearings, citizen advisory boards, focus groups, and surveys (among other activities), all of which are implemented to either determine community priorities, develop policy alternatives, or directly make policy decisions (Yang & Callahan, 2005, p. 199).

Local government managers especially consider engaging the community to be a core competence (International City/County Management Association, 2019; Nalbandian, 1999). An underlying normative argument, Denhardt and Denhardt (2000) stated, “The primary role of the public servant is to help citizens articulate and meet their shared interests rather than to attempt to control or steer society” (p. 549). While social media offer channels for such practices, Nabatchi and Leighninger (2015) point out limitations. For example, individuals may identify public problems or indicate their policy preferences on Facebook and Twitter, but not necessarily engage in more intensive, deliberative conversations. Moreover, anonymous accounts limit transparency and erode people’s trust in social media content, and the digital divide—characterized by insufficient digital literacy and the lack of access to devices and the Internet) restricts inclusion (Feeney & Porumbescu, 2020).

While scholarship reveals a dearth of social media involvement efforts in aggregate (Mergel, 2016; Oliveira & Welch, 2013; Zavattaro & Brainard, 2019), best practices exist to encourage participation and are worth considering. In all, some public managers invest considerable time and resources to promote dialogue. For example, as part of President Obama’s Open Government Initiative, Challenge.gov enabled federal agencies to implement problem-solving contests in which they identified salient problems and defined the goals they hoped to achieve. Participants then (a) submitted ideas through a website, (b) evaluated other people’s submissions, and/or (c) selected their preferred solutions (Mergel & Desouza, 2013). For various reasons, solutions were seldom adopted. Relatedly, Bryer (2013b) identified citizen comments on Regulations.gov “to be emotional, illogical, and lacking in credibility” (p. 263). This limited their value to regulators, and, therefore, few comments were cited as useful for decision making purposes.

More effective perhaps, the Metropolitan Transit Authority (MTA) in Southern California convened dialogue on Facebook about transportation projects that better prepared residents to effectively contribute. Bryer (2013a) describes the MTA administrator as a neutral convener who (a)

posted regular project updates, (b) encouraged assenting and dissenting views, and (c) enforced comment guidelines to maintain a focused, collegial conversation.

Similar approaches relate to emergency management. Sievers (2015) prescribes citizen involvement efforts to generate community-based solutions to disaster planning problems. Moreover, social media may be used to make meaning or sense following a disaster and discuss suggest paths to go forward (Stern, 2017). However, challenges exist in managing related dialogue. Tagliacozzo (2018) points out that after the 2011 New Zealand earthquakes, “dealing with frustrated and distressed residents was even more difficult on social media, because people were inclined to see these platforms as a way to express their dissatisfaction toward the recovery agencies” (p. 7). Relatedly, it is key to note the political challenges of such dialogue. People hold authorities to high standards following disasters (Kapucu & Van Wart, 2006). Depending on operational effectiveness and public perception, organizational, political, and managerial reputations are at stake (Boin et al., 2017). Therefore, some leaders may prefer to refrain from such conversations. For example, Panagiotopoulos et al. (2016) report no such dialogue on Twitter after two extreme events in the United Kingdom in 2010 and 2011, and Wukich (2016) identifies no examples across a series of major disasters in the U.S. from 2013 to 2014. Consequently, decisions to refrain from citizen involvement efforts deny the public a voice and short circuits feedback to officials (Mergel, 2013, 2017).

Methods

Case selection

Disasters create urgent need, require increased interaction between government and community members, and can expose deep-seated community vulnerabilities that lead to suboptimal response and recovery outcomes (Comfort et al., 2010; Tierney et al., 2001). Such focusing events may lead to dialogue and policy change (Birkland, 1997). Hurricane Florence provides a case study that demonstrates such an event on a large scale. The storm made landfall on September 14, 2018, causing considerable damage across North Carolina, South Carolina, and Virginia in both coastal and inland communities. Sixty-two cities, all with populations over 30,000, were selected for the study with the assumption that larger localities possess greater capacity for social media management than smaller ones (Mossberger et al., 2013). With a history of large-scale disasters in the region, perhaps these cities also demonstrate higher levels of disaster preparedness and more sophisticated crisis communication abilities as

compared to those in other areas. Findings may not be reflective of all 50 states and, therefore, not necessarily generalizable in terms of messaging strategies used, but their approaches may represent effective best practices worthy of consideration. Regardless, the case selection offers a baseline set of data from which to draw initial conclusions regarding the types of messaging employed.

Data collection

City government Facebook posts (2,358) provide data for analysis.¹ Far more United States adults use Facebook than Twitter, 69% and 22% respectively (Perrin & Anderson, 2019). During crises, authorities favor Facebook to reach the general public because they can connect with more people (Eriksson & Olsson, 2016). Therefore, Facebook is an appropriate choice for this article. Focusing on the cities' original posts, rather than their replies to public comments, can help to isolate the content that officials prioritize. As Zavattaro and Brainard (2019) point out, social media create an online space, and original posts indicate how public managers seek to populate that space.

The software platform Netlytic collected messages from seven days before the storm made landfall to fourteen days after. The period accounts for a range of disaster management activities (see Comfort & Haase, 2006; Hu et al., 2014; Kapucu, 2006) and allows for enough time to capture initial attempts at reflective dialogue in the form of questions and conversation starters posed by authorities (Wukich, 2016). A focusing event, the hurricane possibly shifted risk reduction to the top of the effected communities' policy agendas (see Birkland, 1997); therefore, it is reasonable to seek out evidence of related conversations before the communities returned to normal discourse.

Data analysis

The dataset includes each message's text, date issued, and a hyperlink to the original post. Content analysis was used to assign a variety of categories or message types to the city Facebook posts (see Table 1). Message types were not mutually exclusive; therefore, a single message could exhibit more than one trait. Drawing from previous research (DePaula et al., 2018; Hughes et al., 2014; Panagiotopoulos et al., 2016; Sutton et al., 2014; Wukich, 2016) and an inductive reading of the data, the codebook accounts for (a) customer-oriented posts about service disruptions and resumption, (b) partner-oriented posts such as guidance on what actions to take and other information that informs resident protective action decisions, and (c)

Table 1. Message types, definitions, and examples.

Message type	Definition and example	Previous studies
Advisories	Guidance on what actions to take to reduce exposure to risk during and after an incident	Sutton et al., 2014; Panagiotopoulos et al., 2016
City Operations	Jacksonville, NC: We can't urge you enough to stay off the roads today. Description of city government activities Greenville, NC: Public Works is working to clean the more than 20,000 storm drains in Greenville as we prepare for #Florence.	Hughes et al., 2014
Citizen Involvement Efforts	Invitation that encourages citizen participation that informs city policymaking Mooresville, NC: Join us at a Public Workshop to help shape the vision and values that will guide development of Mooresville's Comprehensive Plan.	Yang & Callahan, 2007
Conversation Starter	Request for public input or dialogue during nonthreat periods Hampton, VA: What's on your Hurricane checklist?	Wukich, 2016
Event Disruption	Information on delay, cancellation, or rescheduling of an event or meeting Hilton Head, SC: Public Meetings are canceled for the remainder of the week.	Sutton et al., 2014
Favorable Presentation	Description of positive activities to promote likability, competency or worthiness of authorities Wilmington, NC: Still hard at work today Wilmington, we'll keep at it until we're done.	DePaula et al., 2018
Hazard Impact	Information on damage, casualties, and other consequences of an incident	Hughes et al., 2014; Sutton et al., 2014
Intelligence Gathering	Gastonia, NC: A large tree fell into the street and a house on Edgemont Avenue last night. No one was injured. Request for information related to response and recovery operations Chapel Hill, NC: Be our eyes & ears! You can report #HurricaneFlorence issues, like blocked storm drains and downed trees ...	Wukich, 2016
Preparedness	Information on what to expect from a hazard, what supplies to acquire, or how to reduce exposure to risk prior to an incident	Wukich, 2016, 2019; Olson et al., 2019
Resource Provision	Roanoke, VA: Residents are asked to prepare for the storm by gathering emergency supplies: Three-day supply of water and nonperishable food ...	Hughes et al., 2014
Resource Seeking	Information regarding available resources and how to acquire them Charleston, SC: Sandbags will be available tomorrow, while supplies last. Request for volunteers, money, or other resources	Hughes et al., 2014; Wukich, 2016
Service Coproduction Guidance	Myrtle Beach, SC: If you wish to volunteer your services or make a donation during this difficult time, contact the Waccamaw Volunteer Organizations Active in Disasters. Guidance on what actions to take in support of public service provision Fayetteville, NC: Storm Debris Collection to begin next Monday. If you have debris from the storm click the link to learn how to sort your items for pick up ...	—
Service Disruption	Information on delay, cancellation, or resumption of public service provision	Sutton et al., 2014; Panagiotopoulos et al., 2016
Situational Information	Chesapeake, VA: Reminder ... no trash/recycling pickup today or tomorrow. Information on location, nature, and changing conditions of an incident Jacksonville, NC: The roads are very dangerous to travel, covered with lots of debris, large trees, camouflaged power lines, and high water.	Hughes et al., 2014
Warnings	Alerts and indications of danger Sumter, SC: Current forecasts for Hurricane Florence are showing a combination of high wind speeds and the possibility of 12 inches of rainfall ...	Sutton et al., 2014

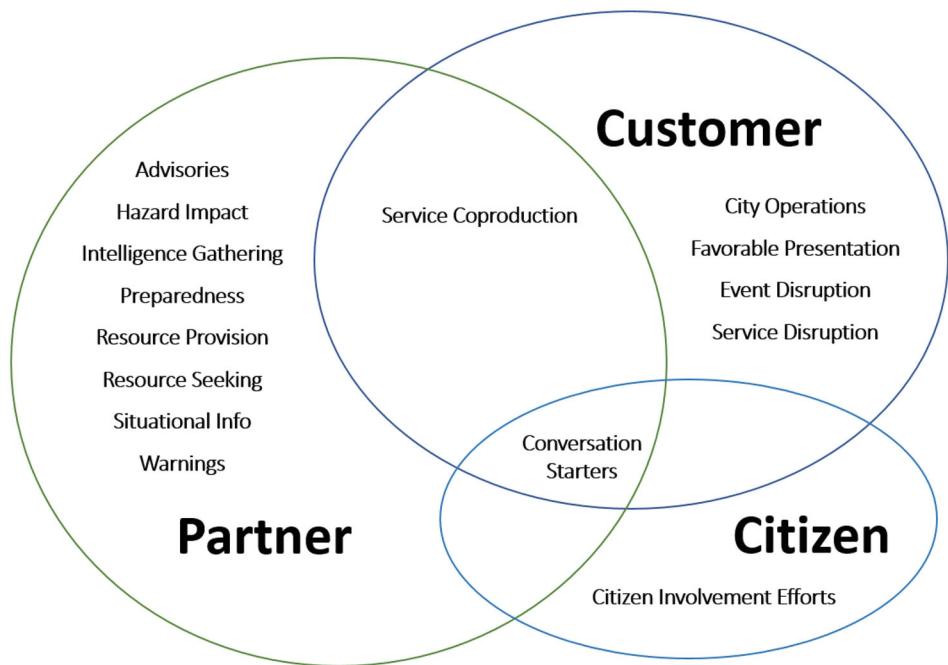


Figure 1. Relationship between public roles and message types.

citizen-oriented prompts to facilitate participation.² Table 1 presents the specific message types, definitions, and examples.

Originally, a team of four researchers reviewed and discussed messages to determine the appropriateness of the codebook. Later, one researcher assigned message types to all posts manually and later applied 50 search terms likely associated with specific categories (e.g., evacuate, rumor, supplies, volunteer, and warning) to spot-check the coding and ensure the accuracy of the overall process. A second researcher coded all the messages for all variables. Intercooder reliability was measured using Cohen’s kappa, and the lowest score was 0.865, which indicates strong agreement across categories.

Based on message type definitions and the context in which they were implemented in this case, the author associated message types to specific public role categories—customers, partners, and citizens. For example, messages that informed the public about disruptions to public service delivery (e.g., service disruption) were associated with a customer orientation. Guidance on what actions to take (e.g., advisories and preparedness) were associated with a partner approach because messages engaged residents in the coproduction of public safety and community resilience. Finally, citizen involvement efforts messages were associated with a citizen orientation because they encouraged participation in the policymaking process.

Message type alignment with public roles were not always mutually exclusive. Figure 1 illustrates that while most message types aligned with

only one public role, message types such as service guidance and conversation starters blended two or more roles. It is understandable to observe such an occasional crossover for Thomas (2013) explains personnel may employ public roles simultaneously.

To analyze message frequency and how public managers enacted and balanced such content, line charts were employed to illustrate the frequency of message types by date across all cities. Only customer and partner-oriented line charts are presented due to the very low frequency of citizen-oriented posts (see Figures 2 and 3). However, citizen-oriented posts are later described in detail. Measures of central tendency and dispersion also were used to indicate the extent to which cities enacted certain practices (see Appendix).

Findings

This section presents the analysis of message types that cities used to disseminate critical information to residents, grouped by public roles—customers, partners, and citizens. First, customer-oriented content is presented that addressed the availability of public goods and services. Next, messages that engaged residents as partners who coproduce public safety and community resilience are described. The section accounts for pre-event mitigation and preparation efforts as well as subsequent response and initial recovery activities. Then illustrated are the few examples of citizen-oriented messages that encouraged residents to consider pertinent policy questions.

Customers

The storm, which made landfall on September 14, 2018, disrupted the ability of many cities, both coastal and inland, to carry on business as usual. They postponed waste collection, suspended transit service, canceled events, and closed city offices. To minimize the inconvenience to their communities, many cities (51) informed residents about such disruptions and then posted further announcements once operations returned to normal ($m = 3.76$, Median = 3, $SD = 3.33$). For example, Chesapeake, Virginia posted, “Reminder... no trash/recycling pickup today or tomorrow.” Hilton Head, South Carolina, posted “Administrative offices, parks and facilities will remain closed... Public Meetings are canceled for the remainder of the week.” Figure 2 demonstrates the frequency of service and event disruption messages. A sharp decline in message frequency occurred the day after the storm (September 15), which suggests that response activities (and related messages) took priority. The subsequent uptick a day later demonstrates cities had effectively dealt with the initial hazard impact and

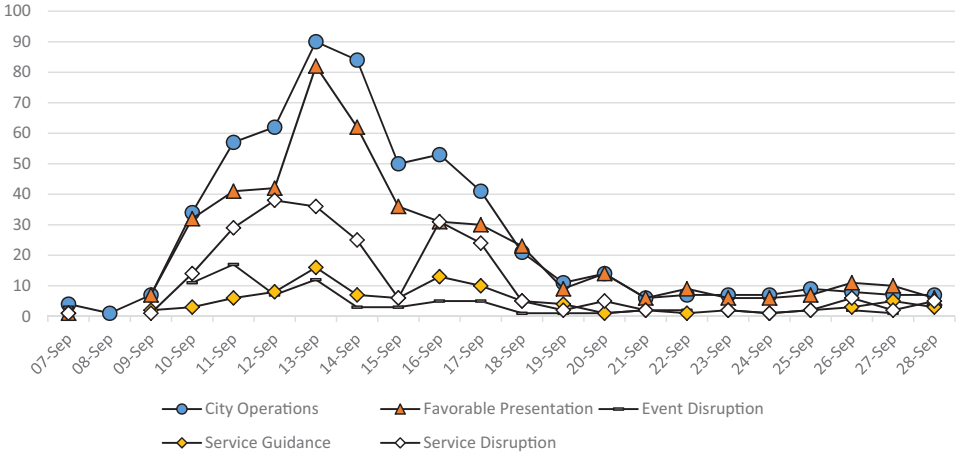


Figure 2. Customer-oriented posts by date.

were able to start scheduling the resumption of normal activities. Wilson, North Carolina posted, “Offices will open on our usual schedule Monday. This includes all recreation centers, administrative offices, and the customer service business office on Nash Street.”

More frequently, cities reported on their activities (see [Figure 2](#)). Raleigh, North Carolina posted, “Between midnight and 7 pm tonight, Raleigh police and firefighters responded to more than 150 Hurricane Florence storm-related calls. Everything from downed trees to malfunctioning traffic lights to electrical hazards.” Fifty-nine cities posted such messages ($m=9.31$, Median = 7.5, $SD=7.59$). This practice provided an element of transparency so that residents were made aware of related problems and how the city worked to address them.

Many messages contained components of favorable presentation. Wilmington, North Carolina posted, “Still hard at work today Wilmington, we’ll keep at it until we’re done. There is at least 650,000 cubic yards of debris in the city to pick up.” Similar messages cast operations in a favorable light and perhaps were employed to reassure residents of the efficacy of their city officials.

Greenville, North Carolina shared a personal moment caught on video between a firefighter and his daughter to dramatize the sacrifice made by first responders. “BYE DAD!: Check out this sweet video of one of our Greenville, N.C. Fire & Rescue firefighters saying goodbye to his daughter as he leaves for work during #HurricaneFlorence for the next several days. Please keep all first responders in thoughts and prayers. #HerHero #FirstResponders.” The message incorporated characteristics of favorable presentation (e.g., #HerHero). The video shows residents first-hand the commitment demonstrated by public servants. [Figure 2](#) illustrates that cities posted most favorable presentation messages in the days leading up to

hurricane's landfall, perhaps to assure the public that personnel were preparing.

Evidence demonstrates some overlap between public roles. Following the storm, several cities engaged residents as partners in the debris removal process, a process that coincided with regular sanitation services. Indian Trail, North Carolina posted:

In order to help quickly clean up streets following this weekend's storm, residents will be able to place more yard waste at the curb than what is typically collected by our solid waste collection provider. Tree limbs and other debris can now measure up to 6 feet in length, and up to 8 inches in diameter. Please place these items by the curb as you typically would with yard waste, and it will be collected on your recycling day.

Thirty-nine cities distributed at least one service coproduction guidance prompt ($m = 1.63$, Median = 1, $SD = 2.15$). Service coproduction guidance contributed to recovery efforts by leveraging regular services (e.g., garbage removal); however, the instructions were out of the ordinary, sometimes complex, and at times required clarification. Presumably, after questions from their residents, Myrtle Beach officials posted, "We heard you! The paper bag deadline for yard debris has been moved back to November 1 ... And the reason for the switch [from plastic bags] is so the Solid Waste Authority can compost more yard waste and save room in the landfill." Again, this message type suggests the blurring lines between customer and partner-oriented approaches.

Partners

In line with the urgency of the situation prior to the storm's landfall, public managers relayed preparedness tips, warnings, and guidance on how to stay safe during the storm. Messages commonly urged people to take action to limit their exposure to risk. Following landfall, hazard impact descriptions, other situational information, and additional advisories (e.g., when to stay home, when to travel, and what areas to avoid) helped to inform residents about how to protect themselves and their loved ones.

Meteorologists predicted the storm a week in advance, and officials used that time to offer preparedness tips on (a) what to expect from the hurricane; (b) what elements to include in a family communication plan (e.g., phone lists and designated meeting locations); and (c) what supplies to acquire (see [Figure 2](#) for the frequency of such messages). High Point, North Carolina posted a common theme, "As our state prepares for #Florence, remember that emergency kits are useful!" They went on to recommend acquiring bottled water, nonperishable food items, batteries, and radios.³

Other content was more interactive in nature. For example, Hampton, Virginia shared a list of supplies and asked, “What’s on your Hurricane checklist? Are we missing anything?” By engaging residents to share ideas, this conversation starter elicited a relatively high number of audience reactions, comments, and shares. Moreover, participants who contributed ideas demonstrated social proof to their virtual neighbors that they were indeed preparing. Such content was limited; only three cities disseminated these types of interactive conversation starters to promote risk reduction concepts.

One-way prompts were far more frequent, and many such messages encouraged residents to engage in flood mitigation efforts prior to the storm, such as clearing storm drain obstructions to reduce the probability of neighborhood flooding. Along the same lines of mitigation, cities offered residents resources such as sand and sandbags to reduce their home’s susceptibility to flooding (see resource provision in [Figure 3](#) and its high frequency in the lead up to the hurricane). Charleston, South Carolina posted, “More than 19,000 sandbags were distributed in the city today. Sandbags will be available tomorrow, while supplies last...” They go on to identify staging locations (e.g., a grocery store, a fire station, and two city government facilities). Messages often included distribution times and locations; others notified people when supplies were no longer available.

As the storm approached, cities increasingly disseminated warnings or indications of danger. The National Weather Service provided a critical source not only for hurricane information, but for related flash flooding and tornado warnings as well. Cities incorporated this information into their content and followed up with advisories or guidance regarding actions for people to take that were by far the most common message type. The advisories included evacuation information when applicable and travel

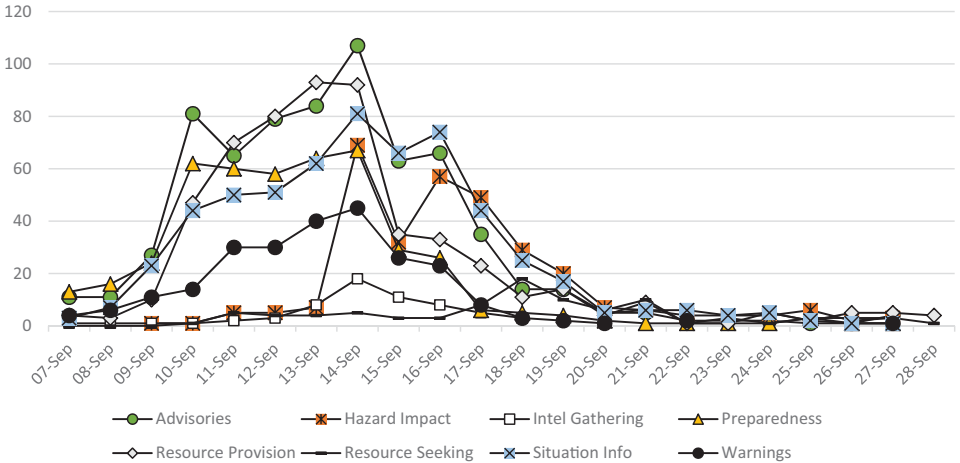


Figure 3. Partner-oriented posts by date.

recommendations such as what areas to avoid. For example, all three states, North Carolina, South Carolina, and Virginia, issued evacuation orders for coastal communities. Hilton Head, South Carolina posted, “Governor Henry McMaster ordered the mandatory evacuation of Beaufort and counties along the coast of South Carolina starting at 12:00 p.m. tomorrow (Tuesday, September 11th). We urge Beaufort County residents and visitors to evacuate ahead of the mandatory time for ease of travel.” They go on to include route information for drivers and public transportation options for those without the means to self-evacuate.

Figure 3 indicates that landfall, September 14, was the busiest day for both warnings and advisories. Many messages described the impending hazard and outlined both what to do and what not to do. Hazard impact and other situational information provided context (see Figure 3). Jacksonville, North Carolina warned residents, “The roads are very dangerous to travel, covered with lots of debris, large trees, camouflaged power lines, and high water.” Often this type of information was included as part of advisories to offer clear justifications that supported protective action guidance. Chapel Hill, North Carolina posted, “Residents are urged to use caution as many roads are flooded and impassable. Public safety officials are once again reminding motorists not to drive through flooded roads. Turn around, don’t drown.” These driving advisories were common.

Some cities solicited actionable feedback (see intelligence gathering in Figure 3). For example, Chapel Hill posted, “Be our eyes & ears! You can report #HurricaneFlorence issues, like blocked storm drains and downed trees, to the Town of Chapel Hill from your smartphone, tablet, or computer. Get the app today!” While cities disseminated intelligence gathering messages relatively infrequently ($m = 1.05$; Median = 1; $SD = 1.40$), more than half of all observed cities (32) posted at least one such request. This indicates an openness to receive such information.⁴

In addition, many cities looked to engage residents as partners face-to-face. Figure 3 illustrates the frequency of resource provision and seeking messages. Fifty-five cities communicated the availability of resources, such as the aforementioned sandbags ($m = 8.84$, Median = 7; $SD = 8.66$). Resource distribution often required face-to-face contact. Conversely, fewer cities (37) requested resources, volunteers, or funds from residents ($m = 1.47$, Median = 1, $SD = 2.22$). Greenville, North Carolina provides an example. As their flood risk subsided, a cresting river threatened a neighboring city. Greenville posted “We are trying to help out our neighbors in Kinston by taking sandbags to them. You can help by dropping off your sandbags at the Public Works Department recycling center, [street address], by 1 p.m. Sunday.”

Citizens

Most information-seeking prompts that cities posted asked for actionable intelligence during the disaster. Requests for information included identifying people in need and the location of flooded areas. Again, thirty-two cities posted related content. Such prompts emphasized the coproduction of public safety and community resilience. Far fewer cities engaged residents in dialogue about policymaking after the hurricane.

Only three cities posted about related citizen involvement efforts. Two cities promoted in-person meetings following the incident to discuss storm-related policy opportunities. For example, Mooresville, North Carolina requested community input “to help shape the vision and values that will guide development of Mooresville’s Comprehensive Plan.” Newport News, Virginia leveraged the aftermath of the hurricane to inform the public of an ongoing stormwater management project. “The public is invited to learn about the Richneck Area Stormwater Project which will improve stormwater quality and alleviate flooding issues” Therefore, city personnel at least tried to inform the public of their new mitigation policies and perhaps invite feedback, which can be considered as first steps in citizen participation.

One message blended aspects of customer service and coproduction with citizen participation. Virginia Beach, Virginia encouraged residents to complete an online survey following the hurricane, “Responses will be used to develop plans for providing additional debris collection services, funnel need requests to potential volunteer agencies and identify locations for potential additional investments in storm water management infrastructure.” The latter point regarding flood mitigation policies indicates Virginia Beach’s receptivity to citizen participation, while the former about service delivery feedback and coordinating with volunteer agencies suggest customer and partner-orientations.

Discussion

This article demonstrates city social media content during Hurricane Florence, categorized by customer, partner, citizen-oriented messages. Whereas city officials were once relegated to managing internal processes, they are increasingly expected, as a core competency, to engage their communities in different ways (International City/County Management Association, 2019; Nalbandian, 1999). Managing the process is essentially about managing relationships and the expectations and the needs of those community members (Feldman & Khademian, 2002; Zavattaro & Brainard, 2019). This pivot from an internal focus to an external one requires a mindset change that broadens the public manager’s perceptions about the

diversity and the appropriateness of their governing relationships and the information they disseminate (Bryer, 2009). Thomas's (2012, 2013) public roles framework illustrates the options available to city managers, and social media applications provide an avenue to build those relationships and develop relevant content.

The application of Thomas's (2012, 2013) public roles framework contributes to scholarship on government social media use. By distinguishing between public roles, scholars can more thoroughly understand the underlying purpose behind content as well as the potential outcomes of such engagement. Therefore, the article contributes a useful conceptual framework by which to analyze government strategies. Past scholarship generally fails to differentiate between such content. Instead, scholars determine the flow of information (Meijer & Thaens, 2013; Mergel, 2013, 2017) or the extent to which interactions occur (Bonsón et al., 2017; Brainard & Edlins, 2015; Kagarise & Zavattaro, 2017). For example, Lovejoy and Saxton (2012) conceptualize *community building* messages as posts that simply promote "interactivity and dialogue" (p. 343) without recognizing their underlying content and related goals. Olson et al. (2019) adopt a similar approach to characterize content from the National Weather Service. Scholars take a risk in making such broad generalizations in that they are unable to systematically differentiate the purpose of government messages and, relatedly, more deliberative exchanges from less substantial interactions.

Ideally, in this case, cities informed their residents' protective action decisions through warnings, advisories, and situational updates. By following instructions, residents perhaps were safer; their houses and apartments more secure. If this were the case, communities coproduced public safety and community resilience along with their city governments. Also common were straight forward, customer-oriented messages that informed residents about service disruptions and event cancelations.

Other messages blended together both customer and partner roles such as posts about amended garbage collection practices for debris removal following the hurricane. In total, such messages perhaps reduced information asymmetries (and the lack of connections) that denied certain community members knowledge and support in previous disasters, for example, communities in New Orleans following Hurricane Katrina that lacked connections with recovery agencies (see Aldrich, 2012).

While information generally flowed in one direction from government to citizen, several cities did request input from residents, and the requests occurred in different contexts. For example, half of the cities requested information from the public about their needs during response and recovery operations (i.e. intelligence gathering). Some engaged residents in conversations about preparedness activities prior to the hurricane making

landfall. Both approaches—intelligence gathering and conversation starters—represent a partner-orientation to the extent that they motivated residents to take protective actions.

Fewer cities solicited input about policies related to rebuilding and strengthening their communities (i.e. citizen involvement efforts), and those that did either invited residents to participate in-person events or simply surveyed residents for ideas—what Nabatchi and Leighninger (2015) refer to as “thin participation”—as opposed to convening more deliberative, group-based online conversations. So even though a focusing event (i.e. the hurricane) created a window for these types of discussions (see Birkland, 1997), this researcher does not see much evidence of them, albeit within a relatively short period of observation.⁵ Therefore, cities missed opportunities to engage residents as citizens, a common finding across different policy domains and service areas in the context of social media (Bryer, 2013a; Hand & Ching, 2011; Mergel, 2013; Zavattaro & Brainard, 2019).

What can practitioners take away from the article? The key for practitioners is to balance and blend public roles appropriately, based in part on the demands of the external environment, especially the needs of the people served. An underlying question public managers should consider is about public value—does content provide substantive value to the community (see Bryson et al., 2014; Nabatchi, 2012; Zavattaro & Brainard, 2019)? Each public role is valuable, but not all are equally valuable, depending on the situation. In urgent, stressful conditions, deliberative conversations can wait until authorities address the more immediate demands of public safety.

This case clearly illustrates different threshold points that signal the appropriateness for certain tactics—preparedness prior to landfall, warnings and guidance (i.e. coproduction) during initial response operations, and customer service throughout. Social media managers should allow for public feedback to inform each practice in real time. Though not frequently implemented in this case, citizen participation has a place both in the aftermath of an incident (Stern, 2017) and throughout more long-term mitigation and preparedness activities (Sievers, 2015). Unfortunately, authorities may prefer to disengage for various reasons. Frustrated residents may vent their grievances as opposed to engaging in dialogue (Tagliacozzo, 2018), and officials risk tarnishing their reputations (Boin et al., 2017). Regardless, they ignore the public at the risk of denying them a voice and short-circuiting valuable feedback loops.

Notably, many cities propagated strategic branding efforts particularly with content designed to cast city operations in a favorable light (i.e. favorable presentation). Improving public perception of service quality represents an underlying goal (see Zavattaro, 2016); therefore, one could argue that these tactics align with a customer orientation to the extent that managers sought to influence

satisfaction with city operations. Social media managers should proceed with caution. Although hard-working city personnel deserve to have their stories told and positive performance recognized, public managers must be careful not to cross the line into propaganda, especially in situations when favorable presentation contradicts the conditions on the ground. The discrepancy between message and reality creates the conditions for public backlash and distrust, particularly during disasters.

Overall, findings suggest practices for cities should be implemented to improve disaster-related social media use by acknowledging all public roles, including:

- Incorporating preparedness information campaigns throughout the year to increase audience awareness and action.
- Making sure people have the information they need to make protective action decisions and the resources to take those actions when needed.
- Encouraging feedback from the public during an incident that informs operational decision-making (i.e. intelligence gathering).
- Advertising opportunities for the public to contribute to larger response and recovery efforts (e.g., volunteer recruitment, service coproduction prompts, fundraising, and other resource seeking messages).
- Providing customer-oriented updates for residents affected by service disruptions.
- Telling the story of how personnel contribute to protecting life and property and how hard they work during these incidents
- Convening dialogue following the incident that allows residents to contribute feedback and ideas that influence policies moving forward.

These evidence-based recommendations may improve operational effectiveness and efficiency while reinforcing democratic norms related to participation.

The article certainly has limitations. The focus on government messages neglects community responses. Each government post, whether inviting comment or not, creates an opportunity for feedback and dialogue. Future research should investigate how community members respond and the public roles they demonstrate either as (a) customers requesting services, (b) partners looking for guidance or offering help, or (c) citizens who state policy preferences for future consideration as well as government responsiveness to those comments. Hughes et al. (2014) point to both the lack of personnel and social media policies that prohibit interactions as possible obstacles. Additional obstacles should be identified that prevent residents from engaging in such conversations. For example, the lack of trust or resident perceptions of insufficient transparency may impede participation (see Song & Lee, 2016), as might more systemic factors such as lack of

access to the Internet or the social and political biases inherent in applications like Facebook and Twitter (Feeney & Porumbescu, 2020).

Other policy domains should be examined. Mergel (2016) reminds us that enactment is highly context-dependent; therefore, this article's findings are not generalizable. During disasters, public managers may be more responsive and open to engage. Urgent need compels them to provide specific types of information (e.g., warnings and advisories) and engage community members in specific ways, such as requesting input to inform operations. In other policy domains, different organizational missions, priorities, and tasks will affect engagement practices (Feeney & Welch, 2016; Mergel, 2012, 2016) as well the extent to which public managers value citizen participation (Bryer, 2011; Yang & Callahan, 2005; Zhang & Feeney, 2018). Those who view themselves as primary authorities in any given field may be less responsive and more likely to reject dialogue (see Bryer, 2009; Yang & Callahan, 2007) whereas more open, inclusive agencies will convene such dialogue. Moreover, the role of social media managers is germane to the extent that they connect residents to relevant information, resources, and agencies within their jurisdictions. Zeemering (2020) analyzes Atlanta, Washington, DC, and San Francisco's Twitter messaging during the COVID-19 pandemic and reports that mayoral staff bridged gaps between what were siloed city agencies in need of coordination. To what extent do social media managers also see their role as connecting residents with city resources? The extent to which they are responsive to the public, help them navigate a fragmented city government landscape, and encourage dialogue may also explain levels of engagement.

Again, social media content is highly context specific based on the policy domain. This complexity contributes to the difficulty scholars experience in creating models that characterize content; hence the reliance on structure (i.e. information flow) to determine the extent that authorities pursue open government goals such as transparency, participation, collaboration. The resulting problem is that scholars too narrowly conceptualize social media content. This article acknowledges the complex, context specific operating conditions of one policy domain, emergency management. While specific results are not generalizable, by demonstrating the relationship between message types and broader public roles, this article contributes a content-based approach that scholars can replicate in other policy domains and service delivery areas.

Conclusion

Thomas's (2012, 2013) public roles model is useful in understanding how public managers frame their social media content specifically to engage the people they serve. The approach accounts for context and contributes to scholarship focused on the structure of information flow. Because social

media enactment depends so much on context, strategies and tactics will likely vary by policy domain and operating conditions. Future research should explore the variation.

Notes

1. These 2,358 posts include 1,851 messages written by the city and 507 shared posts written by other organizations such as state emergency management agencies and utility companies. Whether developed in-house or externally, social media managers assessed this information to be important enough to release publicly. Therefore, it warrants attention in this analysis.
2. A team of four researchers reviewed messages and added service coproduction guidance as a message type, based on their inductive reading of the messages.
3. Cities provided preparedness information—specifically insights regarding associated risks—well after the September 14 landfall (see [Figure 3](#)). This is in part due to the storm's heavy rains and subsequent flooding in many areas that created a more dynamic operational environment.
4. The distinction between intelligence gathering and conversation starters has to do with operational decision-making. Intelligence gathering occurs during response and recovery efforts. Information gathered informs operational decision-making. Therefore, residents participate as partners to improve the services provided in real-time (e.g., emergency response, recovery, public safety, etc.). Conversation starters serve other purposes. Officials may ask questions simply to generate comments—perhaps valuing comments as an engagement statistic or output. They may want to encourage residents to interact with each other to share protective action ideas or create a sense that they are helping each other to get through the incident. Citizen involvement efforts occur when officials convene more deliberative conversations which inform policymaking. In emergency management, policies might include future flood mitigation investments, new disaster plans, or other ways to increase community resilience.
5. Perhaps an extended period of observation may capture more evidence about citizen involvement efforts. However, Wukich (2016) accounted for a year's worth of social media content from U.S. state emergency management agencies and found only a few attempts to convene deliberative conversations despite accounting for several large-scale disasters.

Acknowledgements

The article benefited from several contributions. Kirby Suntala, Gabrielle Parsson, and Alex Mills analyzed message content. Jose Mendez contributed to data analysis and the design of Figure 1 to illustrate the relationship between public roles and message types. The author thanks them as well as the journal's editors and the anonymous reviewers for their guidance. The project was supported by Cleveland State University's Office of Research and the Maxine Goodman Levin College of Urban Affairs.

Notes on contributor

Clayton Wukich is associate professor and MPA director at the Maxine Goodman Levin College of Urban Affairs at Cleveland State University. He studies organizational theory, innovation, and behavior, and his research explores the role of information and communications technology in enabling various forms of participation and collaboration.

ORCID

Clayton Wukich  <http://orcid.org/0000-0001-8359-0402>

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Appendix

Table A1. Message type measures of central tendency and dispersion.

	Cities	%	Mean	Median	SD	Max	Min
Advisories	60	96.8	10.9	8	9.3	36	0
City Operations	59	95.2	9.3	7.5	7.6	29	0
Situational Information	56	90.3	9.2	6	9.5	46	0
Preparedness	55	88.7	7.1	4.5	7.7	42	0
Resource Provision	55	88.7	8.8	7	8.7	40	0
Favorable Presentation	51	82.3	7.5	5	10.7	53	0
Service Disruption	51	82.3	3.8	3	3.3	12	0
Hazard Impact	44	71.0	5.0	3	7.7	41	0
Service Coproduction Guidance	39	62.9	1.6	1	2.2	8	0
Warnings	39	62.9	4.0	2	5.9	32	0
Event Disruption	38	61.3	1.2	1	1.6	7	0
Resource Seeking	37	59.7	1.5	1	2.2	13	0
Intelligence Gathering	32	51.6	1.0	1	1.4	6	0
Conversation Starter	4	6.5	0.1	0	0.4	3	0
Citizen Involvement Efforts	3	4.8	0.1	0	0.3	2	0